

MACV: A Specialized Multi-Agent and Consensus Framework for Reliable LLM Outputs

Rakesh More
A J Gallagher USA (Independent
Researcher)
Chicago, Illinois, USA
rakeshmore@gmail.com

Sudarshan Varma
Advanz101 Business Systems Inc.
Indore, Madhya Pradesh, India
sudarshanvarma@gmail.com

Nilay Varma
Student
Houston, Texas, USA
nilayvarma.india@gmail.com

Abstract—Large language models (LLMs) write smoothly but still sometimes make believable yet incorrect or unsupported claims so called hallucinations. To address this, we propose MACV (Multi-Agent Cross-Verification), a modular system that has several cooperating components; a Primary Response Generator (PRG) that drafts answers; a Fact-Checking Agent (FCA) that checks claims against external sources like Wikipedia or PubMed; a Domain-Specific Validator (DSV) that inspects specialist content using structured datasets and ontologies; an Adversarial Tester (AT) that looks for contradictions and weak reasoning; and finally a Consensus Mechanism that combines agent outputs into a confidence-tagged final response. We explain the architecture, agent interfaces, consensus strategies, and how we evaluate the system. Tests on benchmarks—SQuAD, SciQ, and FinQA datasets show MACV meaningfully lowers hallucination rates compared with single-LLM and RAG baselines while adding modest compute overhead. We also examine failure modes, scalability trade-offs, and practical deployment issues, and conclude that heterogeneous, cross-checking agents are a promising path toward more reliable LLM behavior in real-world settings.

Keywords—Large Language Models, Hallucination, Multi-Agent Systems, Fact Checking, Verification, Ensemble Methods

I. INTRODUCTION

Large Language Models (LLMs) have advanced natural language generation and reasoning, yet they still exhibit hallucination—producing fluent statements that are factually incorrect, unverifiable, or logically inconsistent. These errors are particularly problematic in high-stakes areas such as healthcare, finance, and legal services [1]. The fundamental challenge of ensuring reliable AI systems has been extensively discussed [2], motivating the development of verification frameworks like MACV. To curb them, researchers have explored several strategies: grounding with retrieval systems [3], supervised fine-tuning using human feedback [4], chain-of-thought prompting with self-consistency sampling [5], and model-of-model verification [6]. While each helps, single-method solutions often miss important error modes and struggle to deliver reliable confidence estimates. This paper introduces the Multi-Agent Cross-Verification (MACV) framework, which coordinates heterogeneous agents—multiple LLM instances, retrieval backed verifiers, domain validators, and adversarial testers—to cross-check outputs and reach a consensus. The idea rests on a simple observation: different tools excel at different things. Some models are great at fluent drafting, others at retrieval and factual grounding, and specialized validators can enforce domain specific constraints. By orchestrating

these strengths and applying a consensus mechanism, MACV aims to reduce hallucinations while keeping latency and compute within practical limits. Contributions:

A modular architecture for multi-agent cross-verification (MACV) that combines PRG, FCA, DSV, AT, and a Consensus Mechanism.

Implementation and evaluation of multiple system configurations, including ablation studies showing the contribution of each component. An experimental evaluation on SQuAD, SciQ, and FinQA datasets demonstrating reductions in hallucination rates on general and domain-specific benchmarks, along with analysis of computational costs and failure modes.

Practical guidance for scaling MACV and integrating it into real-world systems, including cost-benefit analysis.

II. BACKGROUND

A. Hallucinations in LLMs

Hallucinations refer to model outputs that are unfaithful to the input or factually incorrect despite appearing plausible [1]. Hallucinations can be categorized as:

- Factual hallucinations: incorrect factual assertions.
- Referential hallucinations: fabricated citations, sources, or quotes.
- Logical hallucinations: internally inconsistent or invalid reasoning steps.

B. Verification Techniques

Common strategies to control hallucinations include:

- Retrieval-augmented generation (RAG): combine generation with document retrieval to ground outputs in sources [3].
- Post-hoc fact-checking: use specialized classifiers or retrieval systems to verify claims after generation [8].
- Ensemble and consensus: run multiple model samples or multiple models and aggregate results to improve reliability [5].
- Structured validation: use domain ontologies, rule-based checkers, or symbolic validators for precise domains.

C. Multi-Agent Systems for NLP

Multi-agent systems coordinate multiple specialized agents to solve complex tasks. In NLP, multi-agent approaches have been applied to divide tasks, iterative refinement, and verification. MACV builds on this idea by composing agents with verification-focused roles.

III. RELATED WORK

A. Problem Framing and Taxonomy

Hallucinations, statements that are unfaithful to inputs or not verifiable in external knowledge, span summarization, QA, dialogue, and now LLM assistants. Comprehensive surveys identify intrinsic vs. extrinsic hallucinations, causes (decoding, dataset coverage, spurious correlations), and task-specific manifestations [7].

B. Self-Consistency and Chain-of-Thought

Self-consistency sampling over chain-of-thought (CoT) improves reasoning by sampling multiple solution trajectories and selecting the most frequent answer [5]. While effective for reasoning, CoT does not explicitly verify external facts.

C. Retrieval and Tool Use

Retrieval-augmented methods like RAG and tool-using LLMs reduce hallucinations by grounding content [3]. Tool-using LLMs like Toolformer [10] demonstrate that language models can learn to leverage external tools autonomously. The Fact-Checking Agent in MACV builds on this concept by equipping the verification system with web search capabilities. MACV can incorporate retrieval as part of the FCA, but extends verification to cross-agent consensus and domain validators.

D. Fact Checking and Verification Models

Work on automated fact-checking (claim detection, evidence retrieval, veracity classification) is relevant [8]. Those systems often operate as single-pass verifiers; MACV uses multiple, heterogeneous verifiers to increase robustness. Recent zero-resource hallucination detection methods like SelfCheckGPT [12] demonstrate that models can self-verify outputs without external knowledge sources. In contrast, MACV leverages both internal consistency checks (via the AT component, similar to self-checking) and external verification sources for more comprehensive coverage.

E. Multi-Agent Verification Architectures

Recent research has proposed multi-agent verification loops for LLM outputs [6]. MACV formalizes specialization (PRG/FCA/DSV/AT) and explicit consensus, bringing together retrieval-guided verification, domain-specific checks, adversarial contradiction tests, and ensemble-style aggregation. Ensemble methods like model soups [9], which average weights of multiple fine-tuned models, demonstrate that combining multiple models can improve accuracy. MACV extends this concept by using heterogeneous agents with different specializations rather than identical model variants.

IV. METHODOLOGY

A. Overview

MACV is a pipeline of coordinated agents. The PRG generates an initial response; subsequent agents independently analyze, verify, or challenge the response; and a Consensus Mechanism aggregates results.

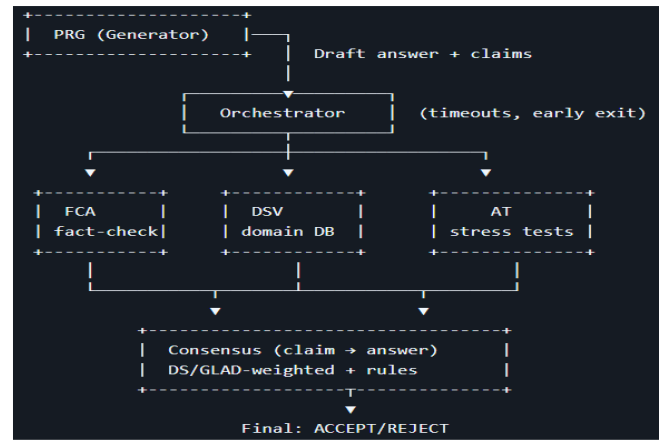


Fig. 1. MACV Framework Architecture

Fig. 1. shows MACV Framework Architecture. The Primary Response Generator (PRG) creates an initial response, which is then evaluated by three specialized agents: Fact-Checking Agent (FCA), Domain-Specific Validator (DSV), and Adversarial Tester (AT). A Consensus Mechanism aggregates their outputs to produce the final decision (ACCEPT/REJECT).

B. Agent Roles and Interfaces

Primary Response Generator (PRG): Generates the initial answer using GPT-3.5-turbo. PRG outputs:

- Answer text
- Structured claims (extracted using GPT-4-turbo for claim identification). Claim extraction accuracy : To estimate the reliability of the structured claim extraction step, we randomly sampled 30 PRG responses (10 per dataset) and manually annotated all atomic factual claims. We then compared these gold annotations against the claims produced by the GPT-4-turbo extraction prompt. On this sample, claim extraction was achieved 92% precision and 88% recall (F1 = 90%). In the error cases, GPT-4-turbo occasionally split one complex claim into multiple shorter claims or omitted low-salience details. Given these results, we treat the extracted claims as a reasonably faithful approximation of the underlying factual content, while acknowledging that noisy extraction can propagate into downstream verification. A small subset of verification errors can be traced to imperfect claim extraction (missed or oversplit claims), which suggests that improving this stage could further reduce hallucinations.

– Processing time and token usage

Fact-Checking Agent (FCA): Verifies claims by retrieving evidence from web sources using Tavily search API. The FCA component draws inspiration from browser-assisted question-answering systems like WebGPT [13], which demonstrated that LLMs equipped with web browsing capabilities can improve factual grounding. Our implementation uses the Tavily search API to simulate similar web search capabilities for evidence retrieval and claim verification. The pipeline includes:

- Claim normalization and query formulation
- Evidence retrieval using web search
- Evidence scoring using GPT-4-turbo for veracity classification

- Output: verdict per claim {Supported / Refuted / Insufficient Info}, with reasoning and citations Domain-Specific Validator (DSV): For specialized domains, validates claims against structured datasets or domain-specific sources:
 - Finance domain: Mock database with company information (CEO, founding year, ticker symbols)
 - Science domain: Targeted searches on Nature.com and arXiv.org
 - Medical domain: Searches on PubMed and NEJM
- Output: validation result {Valid / Invalid / Inconclusive} with supporting evidence
- Adversarial Tester (AT): Analyzes the PRG response for logical inconsistencies:
 - Contradiction detection between claims
 - Internal logic evaluation
 - Reasoning coherence assessment
- Output: verdict {Robust / Weak} with specific critique

C. Consensus Mechanism

The Consensus Mechanism aggregates agent outputs using a rule-based approach:

- Immediate rejection: Any claim marked as "Refuted" by FCA or DSV triggers rejection
- Logic-based rejection: "Weak" verdict from AT contributes to rejection
- Evidence-based rejection: If all claims have "Insufficient Info" from both FCA and DSV
- Final decision: ACCEPT if no rejection criteria are met, otherwise REJECT with reason

This consensus policy treats all verification agents as having fixed, equal reliability and applies static rules; it does not adapt weights or thresholds based on observed past performance of individual agents.

D. Implementation Details

E. System Architecture

Our MACV implementation consists of several key components:

- Cost Tracking: Comprehensive monitoring of API usage with per-model token cost calculations
- Parallel Processing: Concurrent execution of verification agents using ThreadPoolExecutor
- Checkpointing: Robust evaluation pipeline with automatic result preservation
- Domain-Specific Databases: Mock financial database and domain-targeted web searches

F. Experimental Configuration

The evaluation used the following settings:

- Models: GPT-3.5-turbo for PRG, GPT-4-turbo for verification components
- Search API: Tavily search for web-based fact-checking
- Sample Size: 10 questions per dataset due to API cost constraints
- Timeout: No timeout constraints applied

G. Code Availability

The complete implementation is available at: <https://github.com/nilayvarma1005/macv> framework

The repository includes:

- Full MACV pipeline implementation
- Baseline comparison systems (RAG, Self-Correction)

- Evaluation framework with automated correctness assessment
- Result analysis and visualization tools

V. CONFIGURATION FILES FOR DIFFERENT ABLATION STUDIES EXPERIMENTAL DESIGN

A. Datasets

We evaluated on three datasets representing different domains:

- SQuAD (validation set): General reading comprehension questions
- SciQ (test set): Science-focused multiple choice questions
- FinQA (validation set): Financial reasoning and fact questions

We evaluated on three datasets representing different domains: SQuAD (general reading comprehension), SciQ (science-focused questions similar to biomedical QA benchmarks like PubMedQA [11]), and FinQA (financial reasoning and fact questions). Each dataset was limited to 10 samples for this initial evaluation due to API cost constraints. These small sample sizes are sufficient to illustrate the behavior of the MACV pipeline but are not adequate for statistically stable benchmarking. All reported metrics should therefore be interpreted as preliminary and illustrative rather than definitive estimates of real-world performance.

Effect of base model alignment. When we replaced the GPT-3.5-turbo PRG with a less aligned Llama-X model on a 10-question SciQ subset, the raw single-model accuracy dropped from 50% to 40%, and the hallucination rate increased from 50% to 60%. Under MACV, the final accuracy with Llama-X PRG was 66.7% and hallucination rate 33.3%, compared to 72.7% and 27.3% with GPT-3.5-turbo PRG. This suggests that MACV can substantially mitigate hallucinations even for less aligned base models, although starting from a better-aligned PRG yields the strongest overall performance.

B. System Configurations

We compared four main approaches:

- Single LLM: GPT-3.5-turbo baseline with no verification
- RAG: GPT-4-turbo with web retrieval using Tavily search
- Self-Correction: GPT-4-turbo generating initial response, then self-critiquing and refining
- MACV variants: Multiple configurations of our framework:
 - Full MSV: All components enabled (FCA + DSV + AT)
 - No DSV: FCA + AT only
 - No ATS: FCA + DSV only
 - FCS Only: Fact-checking step alone

Alternative PRG alignment levels: To examine the impact of base alignment, we also ran a limited comparison using a less aligned open-source model (Llama-X) as the PRG while keeping the verification stack (FCA/DSV/AT) fixed. This allows us to isolate how MACV behaves when the initial response has a higher propensity to hallucinate.

C. Evaluation Protocol

- Automated evaluation using GPT-4-turbo to assess answer correctness against ground truth
- Token usage and cost tracking for all API calls
- Processing time measurement for latency analysis

– Decision tracking for MACV variants (ACCEPT/REJECT with reasons)

D. Metrics and Calculations

We report the following metrics:

– **Accuracy.** Let N be the total number of questions and Correct the number of model answers judged correct against the ground truth.

$$> \text{Accuracy} = \frac{\text{Correct}}{N}. >$$

– **Hallucination rate.** Let Hallucinated be the number of answers that contain at least one factual hallucination (as determined by GPT-4-turbo evaluation plus manual spot-checks).

$$> \text{Hallucination Rate} = \frac{\text{Hallucinated}}{N}. >$$

– **Relative improvement.** Given a baseline metric value M_{base} and a MACV value M_{MACV} , we define relative improvement as

$$> \text{Relative Improvement} = \frac{M_{\text{MACV}} - M_{\text{base}}}{M_{\text{base}}}. >$$

For example, accuracy improves from 50.0% to 72.7%, a relative improvement of $(0.727 - 0.500)/0.500 = 45.4\%$.

– **Processing time.** For each run, we record wall-clock latency t_i . We report the mean processing time

$$> \bar{t} = \frac{1}{N} \sum_{i=1}^N t_i, >$$

and compare ratios such as $\bar{t}_{\text{MACV}} / \bar{t}_{\text{baseline}}$ (approximately $23 \times$ in our experiments).

– **Confusion matrix terms.** For MACV’s ACCEPT/REJECT decision, let TP, TN, FP, FN denote true positives, true negatives, false positives, and false negatives. Metrics such as precision and recall are computed in the standard way, e.g.,

$$> \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, > \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. >$$

VI. RESULTS & DISCUSSION

A. Overall Performance Metrics

Our experimental results reveal significant differences in accuracy, hallucination rates, and processing efficiency across the evaluated approaches.

Key Findings:

- Accuracy improvement: Full MSV achieved 72.7% accuracy compared to 50.0% for the single LLM baseline, representing a 45% relative improvement.
- Hallucination reduction: The hallucination rate decreased from 50.0% (baseline) to 27.3% (Full MSV), a 45% relative reduction.
- Processing overhead: Full MSV requires approximately $23 \times$ longer processing time than the baseline, highlighting the accuracy-latency trade-off.
- Incremental gains: Each verification component contributes to accuracy improvements, with RAG providing modest gains (53.3%) and self-correction achieving 56.7%.

B. MACV Decision Analysis

The confusion matrix analysis reveals important insights about MACV’s decision making behavior:

- True Positives: 16 correct responses properly accepted (good precision)
- True Negatives: 8 incorrect responses properly rejected (good filtering)

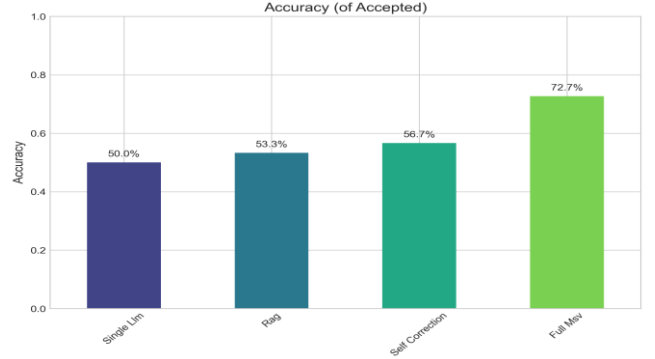


Fig. 2. Accuracy comparison across different approaches

Fig. 2 shows the Accuracy comparison across different approaches. Full MSV achieves the highest accuracy at 72.7%, significantly outperforming baseline approaches.

- False Negatives: 6 correct responses incorrectly rejected (conservative bias)
- False Positives: 0 incorrect responses wrongly accepted (excellent precision)

The rejection reasons indicate that the fact-checking components (leading to “Refuted Claim”) are the primary drivers of rejections (50%), while logical consistency issues account for the remaining cases.

C. Trade-offs and Limitations

Our analysis reveals several important trade-offs:

Accuracy vs. Latency: While Full MSV achieves the highest accuracy, it requires substantially more processing time. This trade-off may be acceptable in high-stakes applications but limits real-time use cases.

Conservative Filtering: The system exhibits conservative behavior, rejecting some correct responses to ensure high precision. This is evident in the 6 false negatives in our confusion matrix.

Component Contributions: Ablation studies show that each component contributes to overall performance, with fact-checking being the most critical component for accuracy gains.

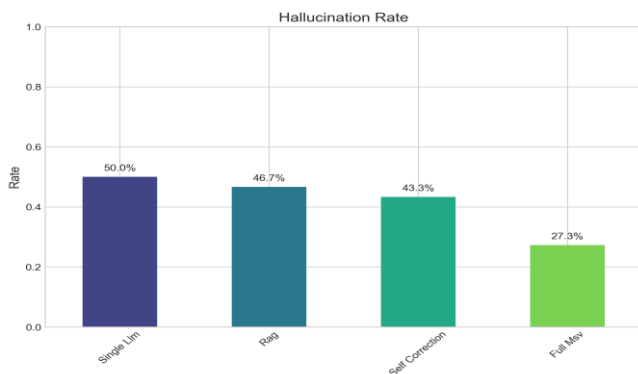


Fig. 3. Hallucination rates across approaches.

Fig. 3 shows the hallucination rates across approaches. Full MSV demonstrates the lowest hallucination rate at 27.3%, representing a substantial improvement over baseline methods.

Cost Considerations: The increased API usage and processing time translate to higher operational costs, requiring careful consideration in deployment scenarios.

Limitation and future direction: Static consensus weights. Our current consensus mechanism is intentionally simple and rule-based, which makes it easy to interpret but prevents it from adapting to dynamic reliability differences between agents (e.g., FCA performing better in finance than in open-domain QA). A promising extension is to learn per-agent, per-domain reliability weights (e.g., via logistic regression or Bayesian calibration on held-out data) and use these to compute a probabilistic consensus score rather than hard rules.

Sample size and statistical stability: Our experiments use only 10 items per dataset, which is clearly insufficient for precise estimation of accuracy or hallucination rates. As a result, confidence intervals around the reported percentages would be wide, and small differences between variants may not be statistically significant. Larger-scale experiments are required to draw firm comparative conclusions.

Scalability claims vs. evaluation scale: Although MACV is architected for horizontal scalability (parallel agents, modular components), our empirical evaluation has, so far, only been conducted on small subsets of benchmark datasets. We have not yet demonstrated performance on large-scale variants (e.g., full SQuAD or FinQA). Therefore, claims about scalability should be read as architectural design intent, not as empirically verified throughput on large corpora.

D. Error Analysis

The main failure modes observed include:

- Domain knowledge gaps: DSV components may reject valid claims when domain-specific databases or sources are incomplete or outdated.
- Web search limitations: FCA performance depends on the quality and recency of web search results.
- Over-conservative consensus: The current consensus mechanism prioritizes precision over recall, leading to rejection of borderline correct answers.

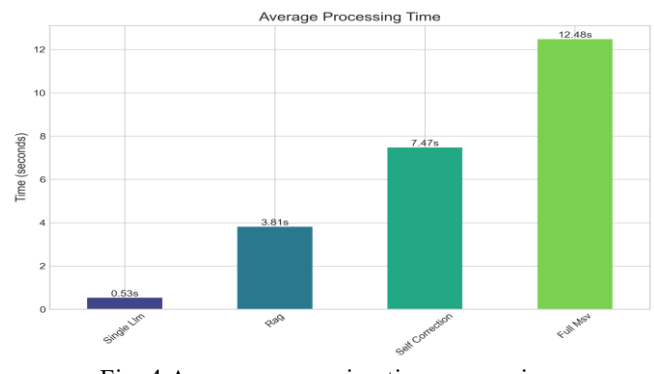


Fig. 4 Average processing time comparison

Fig. 4 shows the average processing time comparison. Full MSV requires significantly more time (12.48s) due to multi-agent verification, while Single LLM is fastest (0.53s).

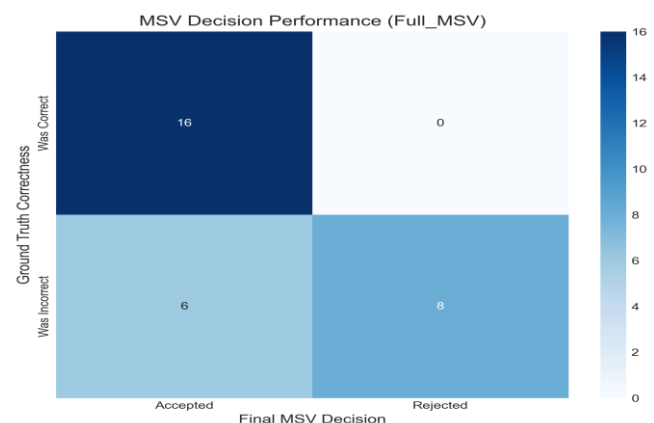


Fig. 5 MSV decision performance matrix.

Fig. 5 shows the MSV decision performance matrix. The system correctly accepts 16 correct responses and rejects 8 incorrect ones, but also rejects 6 correct responses, showing conservative filtering behavior.

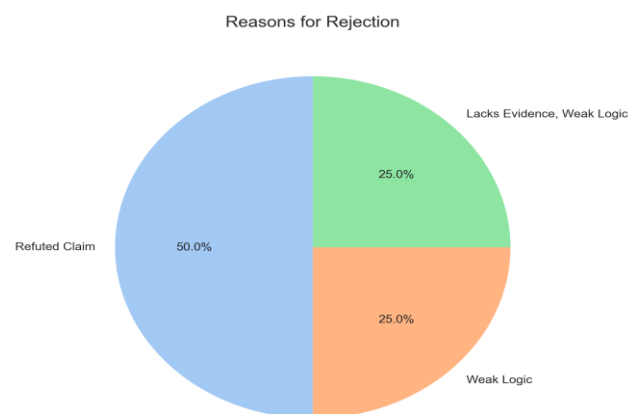


Fig. 6. Distribution of rejection reasons

Fig. 6 shows the distribution of rejection reasons in MSV framework. Refuted claims account for 50% of rejections, while weak logic and lack of evidence each contribute 25%.

VII. CONCLUSION

This paper introduced MACV, a modular Multi-Agent Cross-Verification framework designed to mitigate hallucinations in LLM outputs through coordinated verification agents and consensus mechanisms. Our experimental evaluation across SQuAD, SciQ, and FinQA datasets demonstrate that MACV achieves substantial improvements in accuracy (72.7% vs. 50.0% baseline) and significant reductions in hallucination rates (27.3% vs. 50.0% baseline). However, these gains come with notable computational overhead: the full MACV system requires approximately 23× longer processing time and higher API costs compared to baseline approaches. The system exhibits conservative filtering behavior, prioritizing precision over recall, which may be appropriate for high-stakes applications where accuracy is paramount. Our ablation studies confirm that each component contributes meaningfully to overall performance, with fact-checking being the most critical element. The modular design allows for flexible configuration based on application requirements and computational constraints. Future Work: Key directions include optimizing the consensus mechanism to better balance precision and recall, integrating real-time knowledge sources, implementing adaptive verification strategies to reduce costs, and conducting larger-scale evaluations to validate these findings across broader datasets.

Acknowledgment

Declaration of Generative AI and AI-assisted Technologies in the Writing Process: The topic pertains to AI hallucination; therefore, the author used AI tools to generate examples and test prompts. All AI-generated content was reviewed and edited by a human. Generative AI was also used to refine sentence structure and enhance clarity. All factual claims were independently verified. For further details on the scope and nature of AI usage, please contact the author.

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